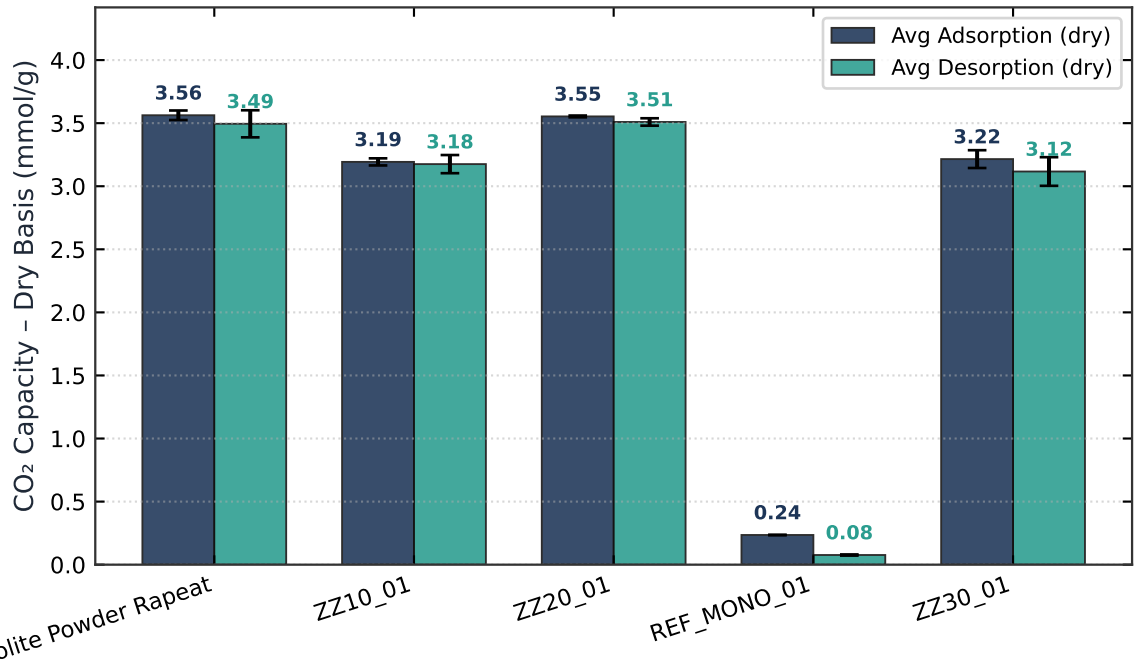
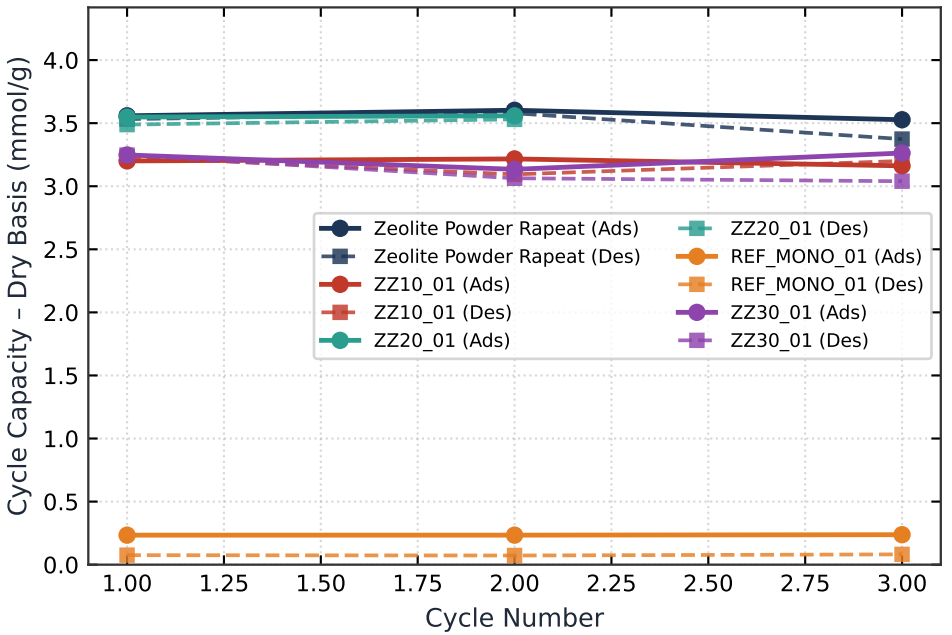
Comparative CO_2 Adsorption & Desorption Analysis Report

Comparative evaluation of 5 selected sample(s) • 15% CO_2 in N_2, 100 mL/min, 24 °C (m/z 44)

Average Adsorption vs. Desorption Capacity



Cycle-by-Cycle Capacity & Regenerability



Comparative Summary Table (Dry Mass Basis & As-Weighed Basis)

Sample / Material	Mass (g)	Dry Mass (g)	Ads C1	Des C1	Avg Ads (Dry)	Avg Des (Dry)	Avg Ads (Wet)	Recovery %
Zeolite Powder Final (Zeolite Powder R	0.6200	0.4873	3.558	3.532	3.563 ± 0.038	3.495 ± 0.107	2.800	98.1 %
ZZ 10 B6 Zeolite (ZZ10_01)	0.3785	0.2975	3.200	3.232	3.193 ± 0.028	3.175 ± 0.072	2.510	99.4 %
ZZ18.5 B6 Zeoite (ZZ20_01)	0.3790	0.2979	3.549	3.489	3.553 ± 0.007	3.509 ± 0.029	2.793	98.8 %
Reference Monolith Valencia	0.9200	0.9200	0.234	0.075	0.235 ± 0.002	0.077 ± 0.005	0.235	32.6 %
ZZ30 Monolith (B6 Zeo) (ZZ30_01)	0.5055	0.3973	3.248	3.247	3.215 ± 0.071	3.117 ± 0.113	2.527	96.9 %